Chapter 5

TCP/IP

Chapter 5 Outline

5.1 - Introduction

5.2 - Transport & Network Layer Protocols

- TCP/IP

5.3 Transport Layer Functions

- Linking to the Application Layer
- Segmenting
- Session management

5.4 - Addressing

- Assigning addresses and address resolution

5.5 - Routing

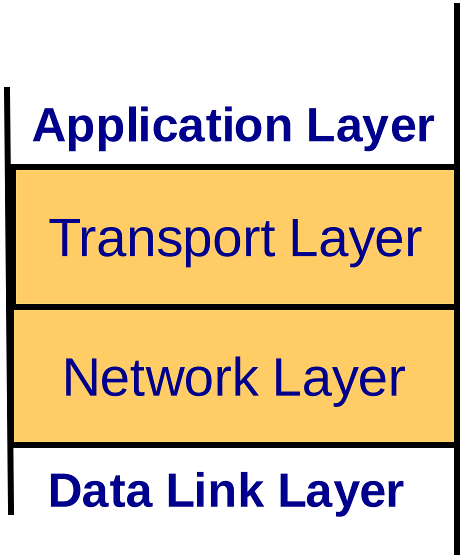
- Types of routing, routing protocols, multicasting, and router anatomy

5.6 - TCP/IP Example

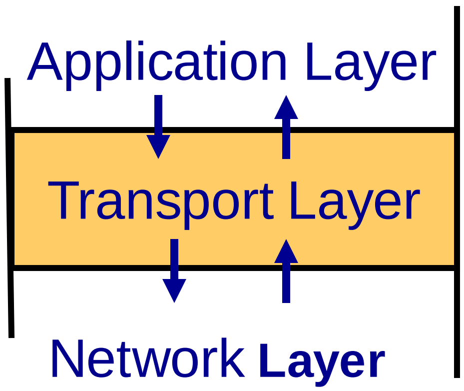
5.7 – Implications for Management

5.1 Introduction

- Transport and Network layers
  - Responsible for moving messages from end-to-end in a network
  - Closely tied together
  - TCP/IP: most commonly used protocol
    - Used in Internet
    - Compatible with a variety of Application Layer protocols as well as with many Data Link Layer protocols



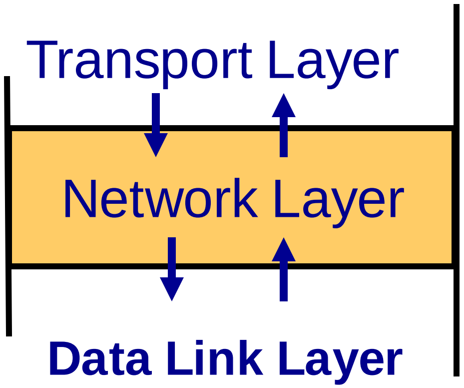
Introduction - Transport layer



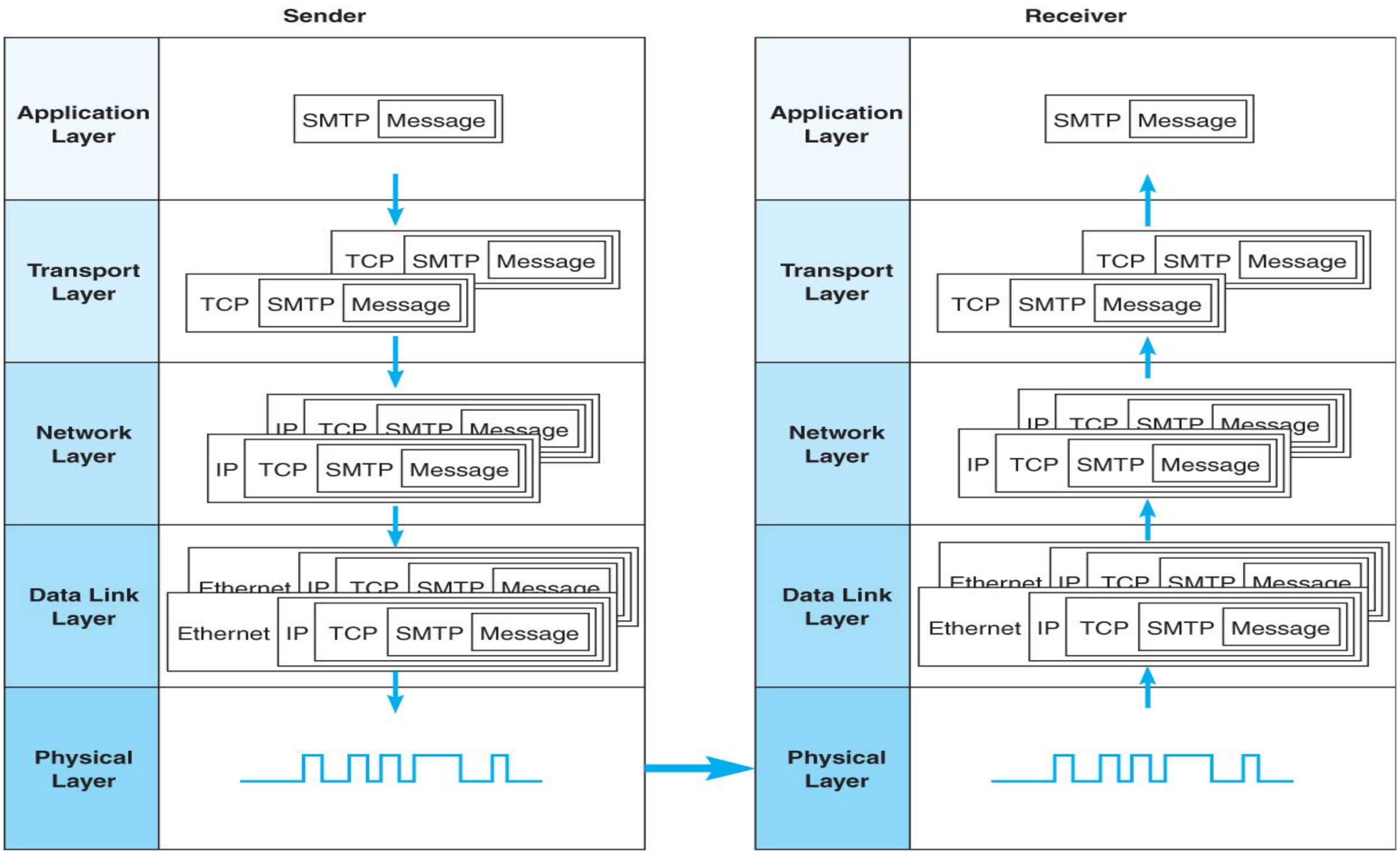
- Responsible for end-to-end delivery of messages
  - Sets up virtual circuits (when needed)
- Responsible for segmentation and reassembly
  - Breaking the message into several smaller pieces at the sending end
  - Reconstructing the original message into a single whole at the receiving end
- Interacts with Application Layer

Introduction – Network Layer

- **Responsible for addressing and routing of messages**
  - Selects the best path from computer to computer until the message reaches destination
- **Performs encapsulation on sending end**
  - Adds network layer header to message segments
- **Performs decapsulation on receiving end**
  - Removes the network layer header at receiving end and passes them up to the transport layer



TCP/IP's 5-Layer Network Model



5.2 Transport/Network Layer Protocols

- TCP/IP (Transmission Control Protocol / Internet Protocol)
  - Most common, used by all Internet equipment
- Developed in 1974 by V. Cerf and B. Kahn
  - As part of Arpanet (U.S. Department of Defense)
- Most common protocol suite
  - Used by the Internet
  - Largest percentage of all backbone, metropolitan, and wide area networks use TCP/IP
  - Most commonly used protocol on LANs
- Reasonably efficient and error free transmission
  - Performs error checking
  - Transmits large files with end-to-end delivery assurance
  - Compatible with a variety of data link layer protocols

Transmission Control Protocol

- Links the application layer to the network layer
- Performs packetization and reassembly
  - Breaks up a large message into smaller packets
  - Numbers the packets
  - Reassembles the packets at the destination end
- Ensures reliable delivery of packets

Source port	Destination port	Sequence number	ACK number	Header length	Unused	Flags	Flow control	CRC-16	Urgent pointer	Options	User data
16 bits	16 bits	32 bits	32 bits	4 bits	3 bits	9 bits	16 bits	16 bits	16 bits	32 bits	Varies

Internet Protocol (IP)

- **Responsible for addressing and routing of packets**
- **Two versions in current in use**
  - **IPv4: a 192 bit (24 byte) header, uses 32 bit addresses.**
  - **IPv6: Mainly developed to increase IP address space due to the huge growth in Internet usage (128 bit addresses)**
- **Both versions have a variable length data field**
  - **Max size depends on the data link layer protocol.**
  - **e.g., Ethernet's max message size is 1,492 bytes, so max size of TCP message field:**

$$1492 - \underbrace{24}_{\text{TCP header}} - \underbrace{24}_{\text{IPv4 header}} = 1444 \text{ bytes}$$

IP Packet Formats

IPv4 Header: 192 bits (24 bytes)

Version number	Header length	Type of service	Total length	Identifiers	Flags	Packet offset	Hop limit	Protocol	CRC 16	Source address	Destination address	Options	User data
4 bits	4 bits	8 bits	16 bits	16 bits	3 bits	13 bits	8 bits	8 bits	16 bits	32 bits	32 bits	32 bits	Varies

IPv6 Header: 320 bits (40 bytes)

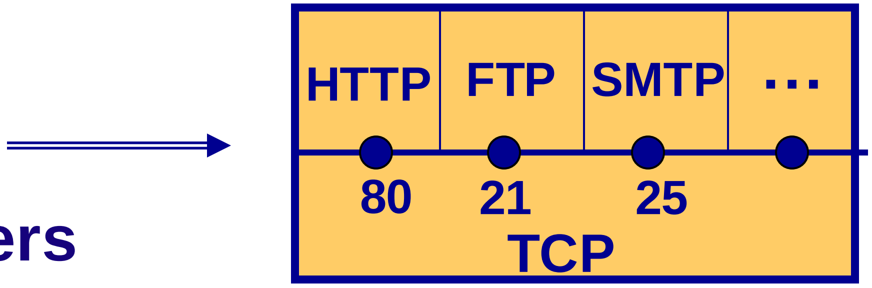
Version number	Priority	Flow name	Total length	Next header	Hop limit	Source address	Destination address	User data
4 bits	4 bits	24 bits	16 bits	8 bits	8 bits	128 bits	128 bits	Varies

5.3 Transport Layer Functions

- Linking to the Application Layer
- Segmenting
- Session management
  - Connection-oriented
  - Connectionless
  - Quality of Service (QoS)

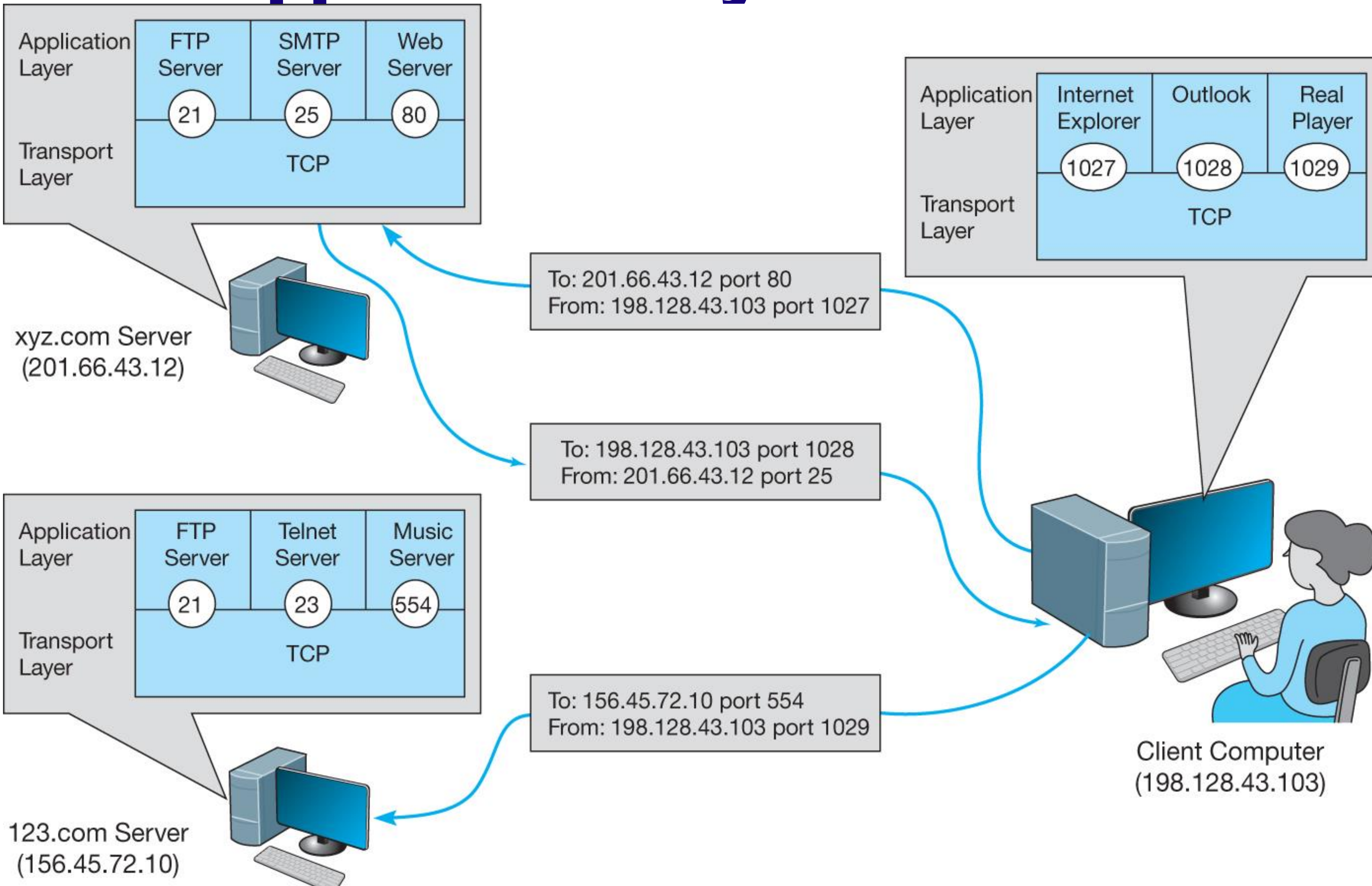
Linking to Application Layer

- TCP may serve several Application Layer protocols at the same time
  - Problem: Which application layer program to send a message to?
  - Solution: Port numbers located in TCP header fields; 2-byte each (source, destination)
- Standard port numbers
  - Usual practice numbers



- Nonstandard port numbers
  - Possible, but requires configuration of TCP
  - Can be used to enhance security from commonly known ports

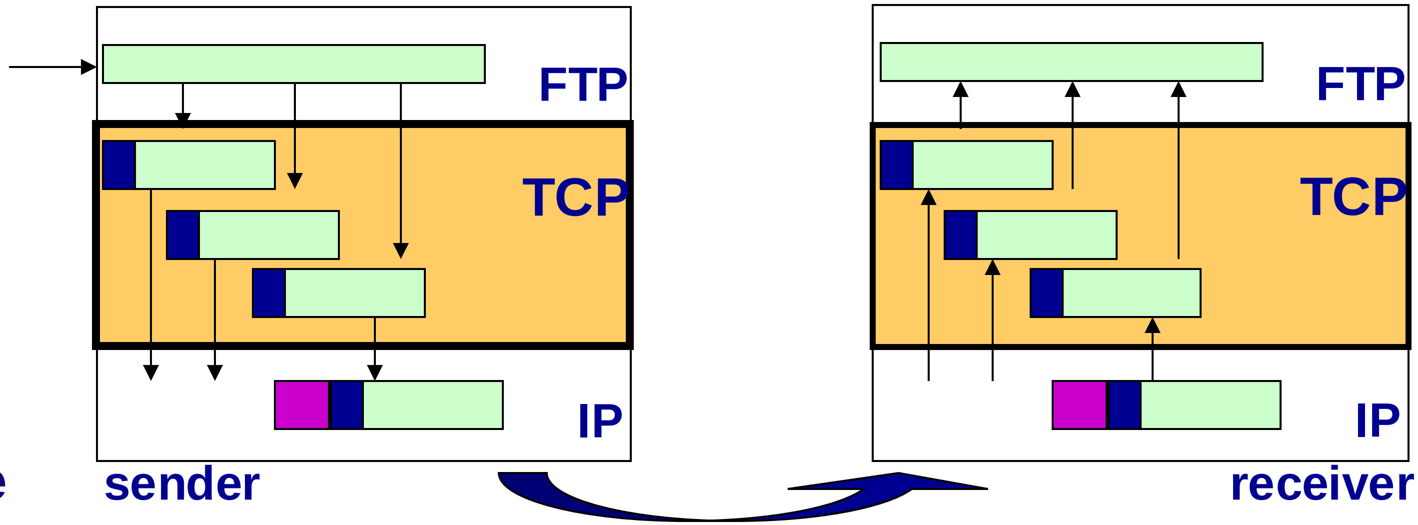
Application Layer Services



Port #	Application	Port#	Application
20/TCP 21/TCP	FTP – data FTP control (command)	389	Lightweight Directory Access Protocol (LDAP)
22/TCP, UDP	Secure Shell (SSH)—file transfers (scp, sftp)	554	Real Time Streaming Protocol (RTSP)
23/TCP	Telnet	587	e-mail message submission (SMTP)
25/TCP	SMTP	636	LDAPS (over TLS/SSL)
53	Domain Name System (DNS)	706/TCP	SILC, Secure Internet Live Conferencing
80/TCP	HTTP	901	VMware Virtual Infrastructure Client
109/TCP 110/TCP	POP2 POP3	8080 /TCP	HTTP alternate; Web proxy and caching; Apache Tomcat
194/UDP	IRC (Internet Relay Chat)	443/TCP, UDP	Hypertext Transfer Protocol over TLS/SSL (HTTPS)
213	IPX		

Packetization and Reassembly

Application layer sees message as a single block of data



TCP breaks a large message into smaller pieces (packetization)

TCP puts packets back together at the destination (reassembly)

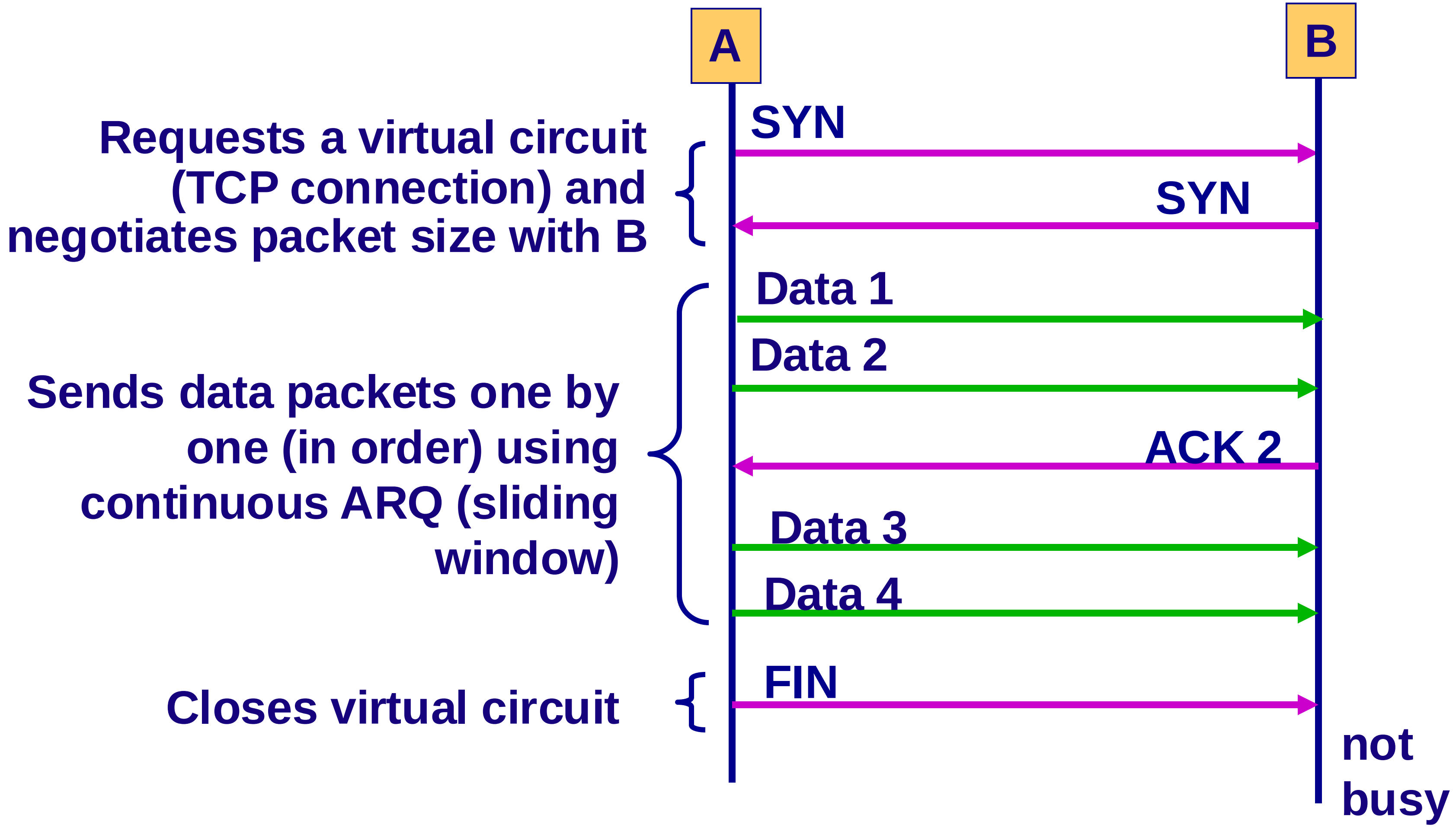
What size packet to use? Done through negotiations

Delivers incoming packets as they arrive (e.g., Web pages) or to wait until entire message arrives (e.g., e-mail)

Session Management

- *Connection Oriented* is provided by TCP
  - Setting up a virtual circuit, or a TCP connection
    - TCP asks IP to route all packets in a message by using the same path (from source to destination)
    - Packet deliveries are acknowledged
    - Used by HTTP, SMTP, FTP
- *Connectionless Routing* is provided by UDP
  - Sending packets individually without a virtual circuit
  - Each packet is sent independently of one another, and will be routed separately, following different routes and arriving at different times
- QoS Routing (provided by RTP)
  - A special kind connection oriented routing with priorities

Setting up Virtual Connections



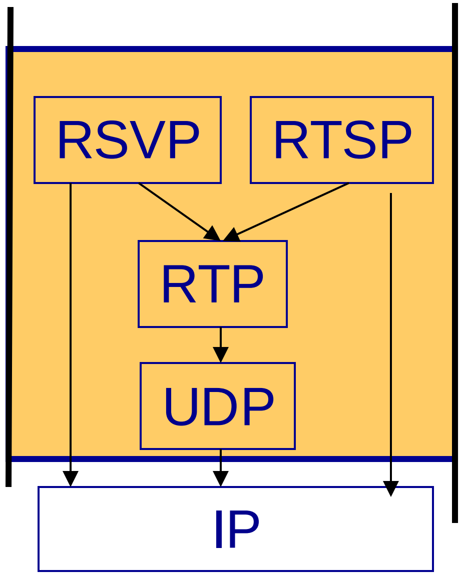
UDP - User Datagram Protocol

- Protocol used for connectionless routing in TCP/IP suite that uses no acks, no flow control
- Uses only a small packet header
  - Only 8 bytes containing only 4 fields:
    - Source port
    - Destination port
    - Message length
    - Header checksum
- Commonly used for control messages that are usually small, such as DNS, DHCP, RIP and SNMP.
- Can also be used for applications where a packet can be lost, such as information rich video

QoS - Quality of Service

- QoS defines and assigns priorities to “classes of service”
- Timeliness - timely delivery of packets
  - Packets be delivered within a certain period of time (to produce a smooth, continuous output
  - Required by some applications, especially real time applications (e.g., voice and video frames)
  - (e-mail doesn’t require this)
- QoS routing
  - Defines classes of service, each with a different priority:
    - Real-time applications such as VoIP- highest
    - A graphical file for a Web page - a lower priority
    - E-mail - lowest (can wait a long time before delivery)

Protocols Supporting QoS



- Asynchronous Transfer Mode (ATM)
  - A high-speed data link layer protocol
- TCP/IP protocol suite
  - Resource Reservation Protocol (RSVP)
    - Sets up virtual circuits for general purpose real-time applications
  - Real-Time Streaming Protocol (RTSP)
    - Sets up virtual circuits for audio-video applications
  - Real-Time Transport Protocol (RTP)
    - Used after a virtual connection setup by RSVP or RTSP
    - Adds a sequence number and a timestamp for helping applications to synchronize delivery
    - Uses UDP (because of its small header) as transport

Network Layer Functions

- Addressing
  - Each device on the path between source and destination must have an address
  - Internet Addresses
  - Assignment of addresses
  - Translation between network layer addresses and other addresses (address resolution)
- Routing
  - Process of deciding what path a packet must take to reach destination
  - Routing protocols

Types of Addresses

Address Type	Example	Example Address	Analogy
Application Layer	URL	www.manhattan.edu	Name
Network Layer	IP address	149.61.10.22 (4 bytes)	Zip code
Data Link Layer	MAC address	00-0C-00-F5-03-5A (6 bytes)	Street addr.

- These addresses must be translated from one type to another for a message to travel from sender to receiver.
- This translation process is called address resolution.
- It is like sending a letter via postal mail to John Smith. The zip code (IP address) is used to get the letter to the city (LAN), and then the street address (MAC) is used to get it to the specific house (computer).

Assignment of Addresses

- Application Layer address (URL)
  - For servers only (clients don't need it)
  - Assigned by network managers and placed in configuration files.
  - Some servers may have several application layer addresses
- Network Layer Address (IP address)
  - Assigned by network managers, or by programs such as DHCP, and placed in configuration files
  - Every network on the Internet is assigned a range of possible IP addresses for use on its network
- Data Link Layer Address (MAC address)
  - Unique hardware addresses placed on network interface cards by their manufacturers (based on a standardized scheme)
- Servers have permanent addresses, clients usually do not

Internet Addresses

- Managed by ICANN
  - Internet Corporation for Assigned Names and Numbers
  - Manages the assignment of both IP and application layer name space (domain names)
    - Both assigned at the same time and in groups
    - Manages some domains directly (e.g., .com, .org, .net) and
    - Authorizes private companies to become domain name registrars as well
- Example: Indiana University
  - URLs that end in .indiana.edu and iu.edu
  - IP addresses in the 129.79.x.x range (where x is any number between 0 and 255)

IPv4 Addresses

- 4 byte (32 bit) addresses
  - Strings of 32 binary bits
- Dotted decimal notation
  - Used to make IP addresses easier to understand for human readers
  - Breaks the address into four bytes and writes the digital equivalent for each byte
- Example: 128.192.56.1

1 0 0 0 0 0 0 0	1 1 0 0 0 0 0 0	0 0 1 1 1 0 0 0	0 0 0 0 0 0 0 1

Classfull Adressing

Class	IP Address Range	Classful Description	Slash Notation	Number of Hosts
A	10.0.0.0–10.255.255.255	One Class A address	10.0.0.0/8	16,777,216
B	172.16.0.0.–172.31.255.255	16 Class B addresses	172.16.0.0/16	1,048,576
C	192.168.0.0–192.168.255.255	256 Class C addresses	192.168.0.0/24	65,536

Class	First byte	Byte allocation	Start Address	End Address	Number of Networks	Number of Hosts
A	1–126	Network.Host.Host.Host	1.0.0.0	126.255.255.255	128 (2^7)	16,777,216 (2^{24})
B	128–191	Network.Network.Host.Host	128.0.0.0	191.255.255.255	16,384 (2^{14})	65,536 (2^{16})
C	192–223	Network.Network.Network.Host	192.0.0.0	223.255.255.255	2,097,152 (2^{21})	256 (2^8)

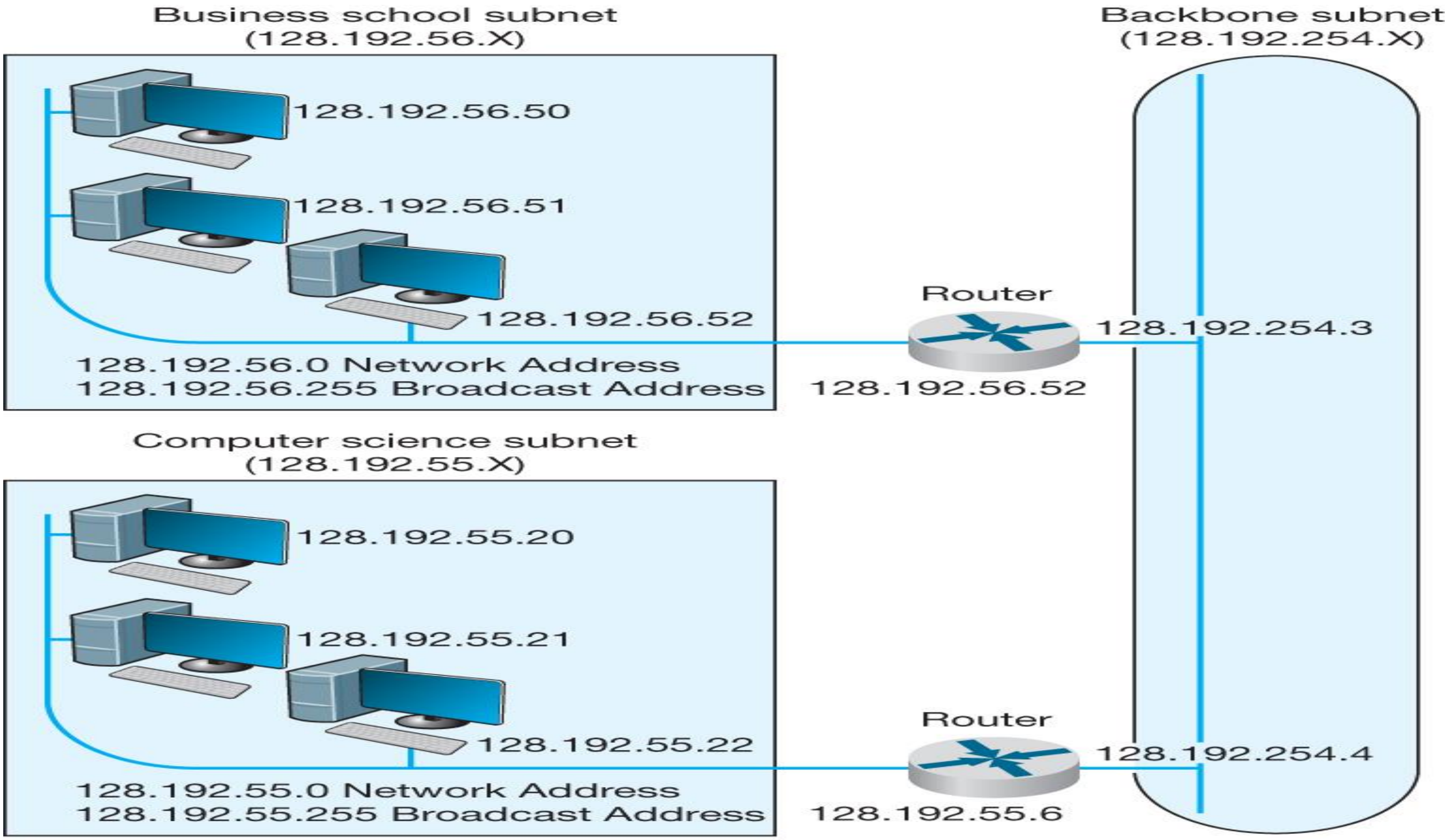
IPv6 Addressing

- Need
  - IPv4 uses 4 byte addresses:
    - Total of one billion possible addresses
  - IP addresses often assigned in (large) groups
    - Giving out many numbers at a time
    - → IPv4 address space has been used up quickly
    - e.g., Indiana University: uses a Class A IP address space (65,000 addresses; many more than needed)
- IPv6 uses 16 byte addresses:
  - 3.2×10^{38} addresses, a very large number
  - Little chance this address space will ever be used up

Subnets

- Group of computers on the same LAN with IP numbers using the same prefix
- Assigned addresses that are 8 bits in length
  - For example:
    - Subnet 149.61.10.x
      - Computers in Business (x is between 0 & 255)
    - Subnet 149.61.15.x
      - Computers in CS department
- Assigned addresses could be more or less than eight bits in length
  - For example: If 7 bits used for a subnet
    - Subnet 1: 149.61.10.1-128
    - Subnet 2: 149.61.10.129-255

Subnets: Example



Subnet Masks

- Used to make it easier to separate the subnet part of the address from the host part.
- Example
  - Subnet: 149.61.10.x
  - 149.61.10.0 is network address. 149.61.10.255 is a broadcast address.
  - Subnet mask: 255.255.255.000 or in binary

    11111111.11111111.11111111.00000000
- Example
  - Subnets: 149.61.10.1-128
  - 149.61.10.0 is network address. 149.61.10.128 is a broadcast address.
  - Subnet mask 255.255.255.128 or, in binary:

    11111111.11111111.11111111.10000000

Dynamic Addressing

- Giving addresses to clients (automatically) only when they are logged in to a network
  - Eliminates permanent addresses to clients
  - When the computer is moved to another location, its new IP address is assigned automatically
  - Makes efficient use of IP address space
  - Example:
    - A small ISP with several thousands subscribers
    - Might only need to assign 500 IP addresses to clients at any one time
- Uses a server to supply IP addresses to computers whenever the computers connect to network

Programs for Dynamic Addressing

- Bootstrap Protocol (bootp)
- Dynamic Host Control Protocol (DHCP)
- Different approaches, but same basic operations:
  - A program residing in a client establishes connection to bootp or DHCP server
  - A client broadcasts a message requesting an IP address (when it is turned on and connected)
  - Server (maintaining IP address pool) responds with a message containing IP address (and its subnet mask)
  - IP addresses can also be assigned with a time limit (leased IP addresses)
    - When expires, client must send a new request

Address Resolution

- Server Name Resolution
  - Translating destination host's domain name to its corresponding IP address
  - www.yahoo.com is resolved to → 204.71.200.74
  - Uses one or more Domain Name Service (DNS) servers to resolve the address
- Data Link Layer Address Resolution
  - Identifying the MAC address of the next node (that packet must be forwarded)
  - Uses Address Resolution Protocol (ARP)

DNS - Domain Name Service

- Used to determine IP address for a given URL
- Provided through a group of name servers
  - Databases containing directories of domain names and their corresponding IP addresses
- Large organizations maintain their own name servers
  - smaller organizations rely on name servers provided by their ISPs
- When a domain name is registered, IP address of the DNS server must be provided to registrar for all URLs in this domain
  - Example: Domain name: *indiana.edu*

    URLs: *www.indiana.edu, www.kelly.indiana.edu, abc.indiana.edu*

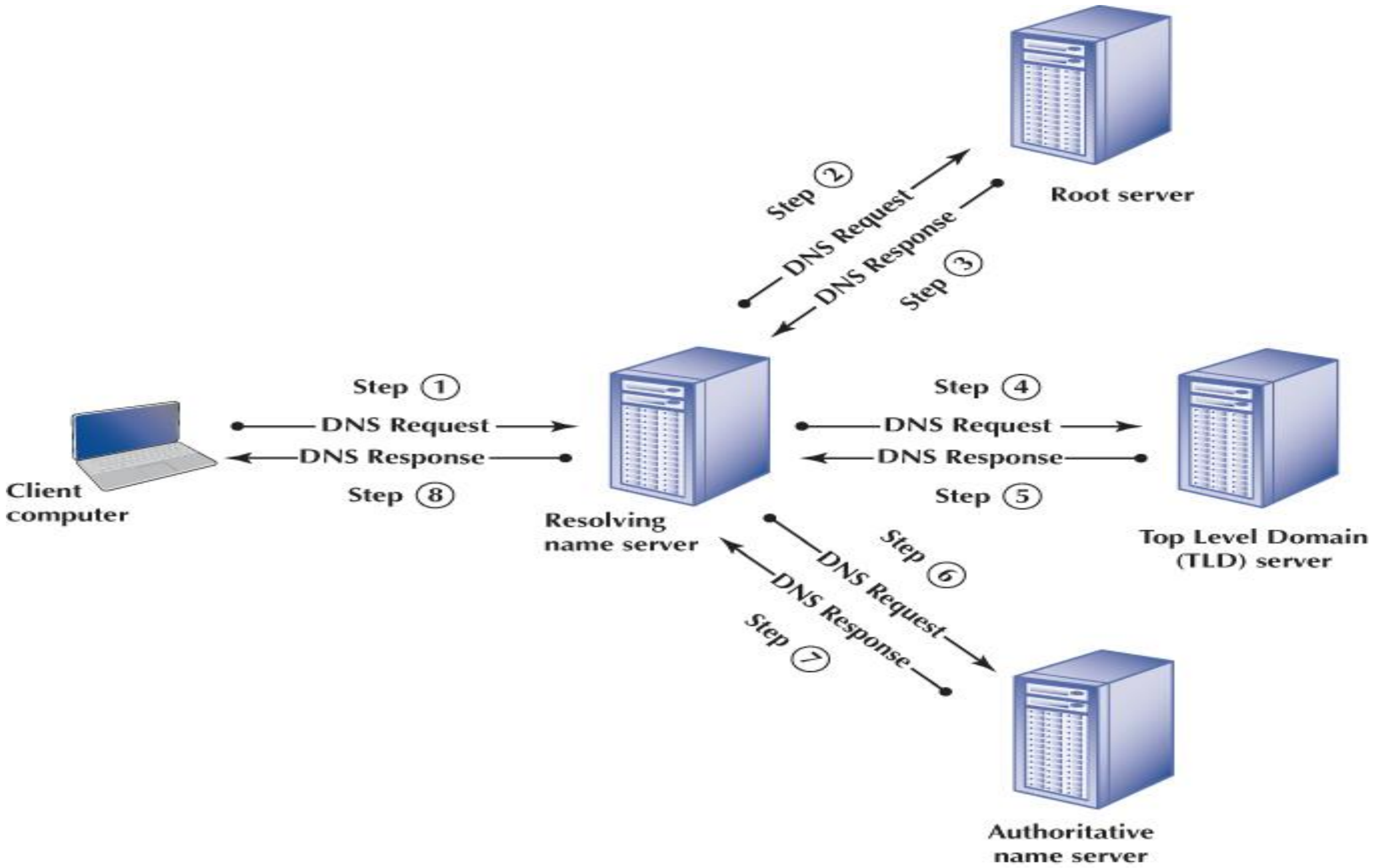
How DNS Works

- If the desired URL is in the client's address table:
  - Use the corresponding IP address
  - Each client maintains a server address table
    - containing URLs used and corresponding IP addresses
- If the desired URL is not in the client's address table:
  - Use DNS to resolve the address
  - Sends a DNS request packet to its local DNS server
  - URL in Local DNS server
    - Responds by sending a DNS response packet back to the client

How DNS Works (Cont.)

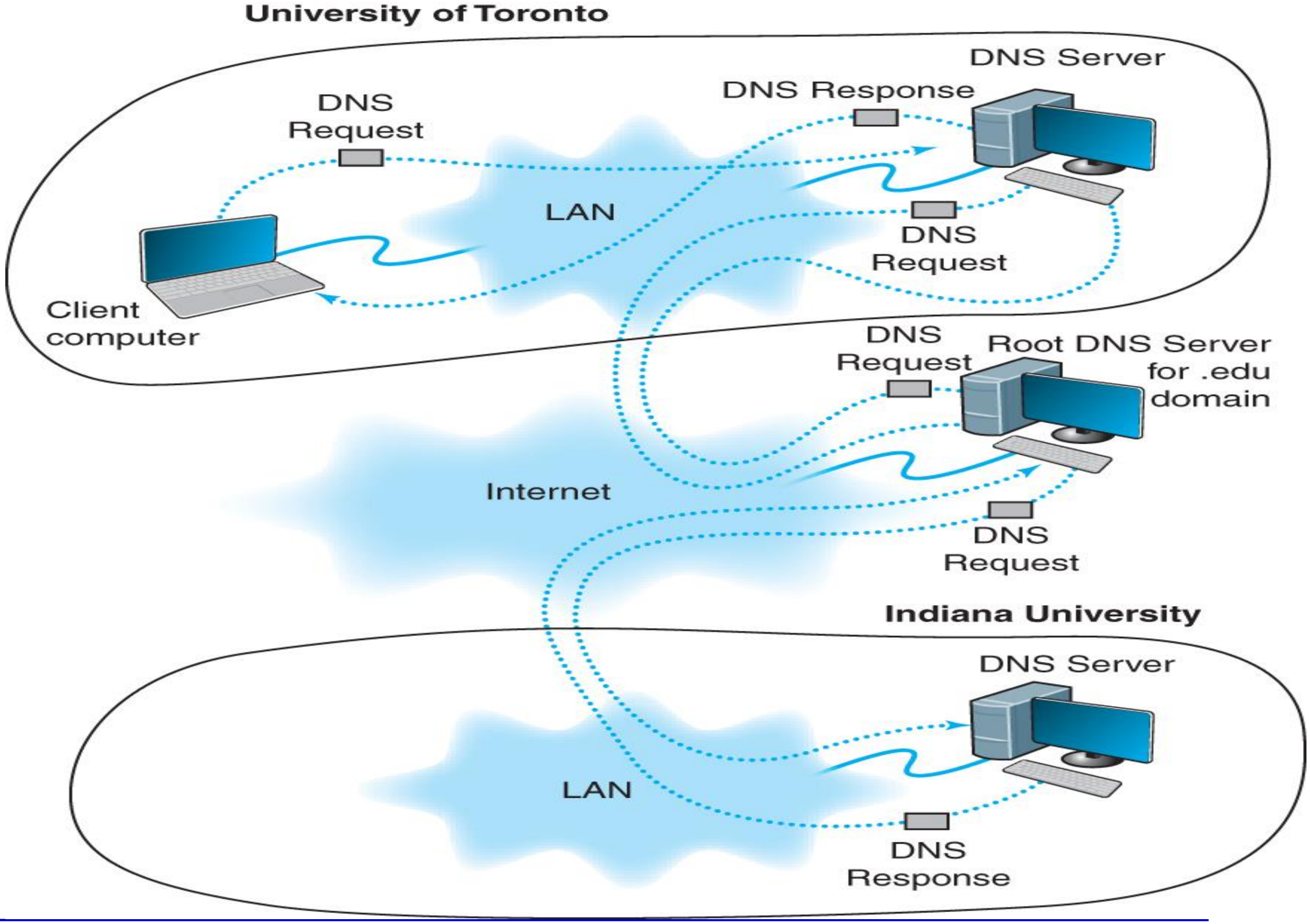
- If the URL is NOT in the local DNS server
  - Sends DNS request packet to the next highest name server in the DNS hierarchy
  - Usually the DNS server at the top level domain (such as the DNS server for all .edu domains)
  - If the URL is NOT in the name server
    - Sends DNS request packet ahead to name server at the next lower level of the DNS hierarchy

How DNS Works



How DNS Works

If client at Toronto asks for a web page on Indiana University's server:



MAC Address Resolution

- **Problem:**
  - Unknown MAC address of the next node (whose IP address is known)
- **Solution:**
  - Uses Address Resolution Protocol (ARP)
- **Operation**
  - Broadcast an ARP message to all nodes on a LAN asking which node has a certain IP address
  - Host with that IP address then responds by sending back its MAC address
  - Store this MAC address in its address table
  - Send the message to the destination node
  - Example of a MAC address: 00-0C-00-F5-03-5A

5.5 Routing

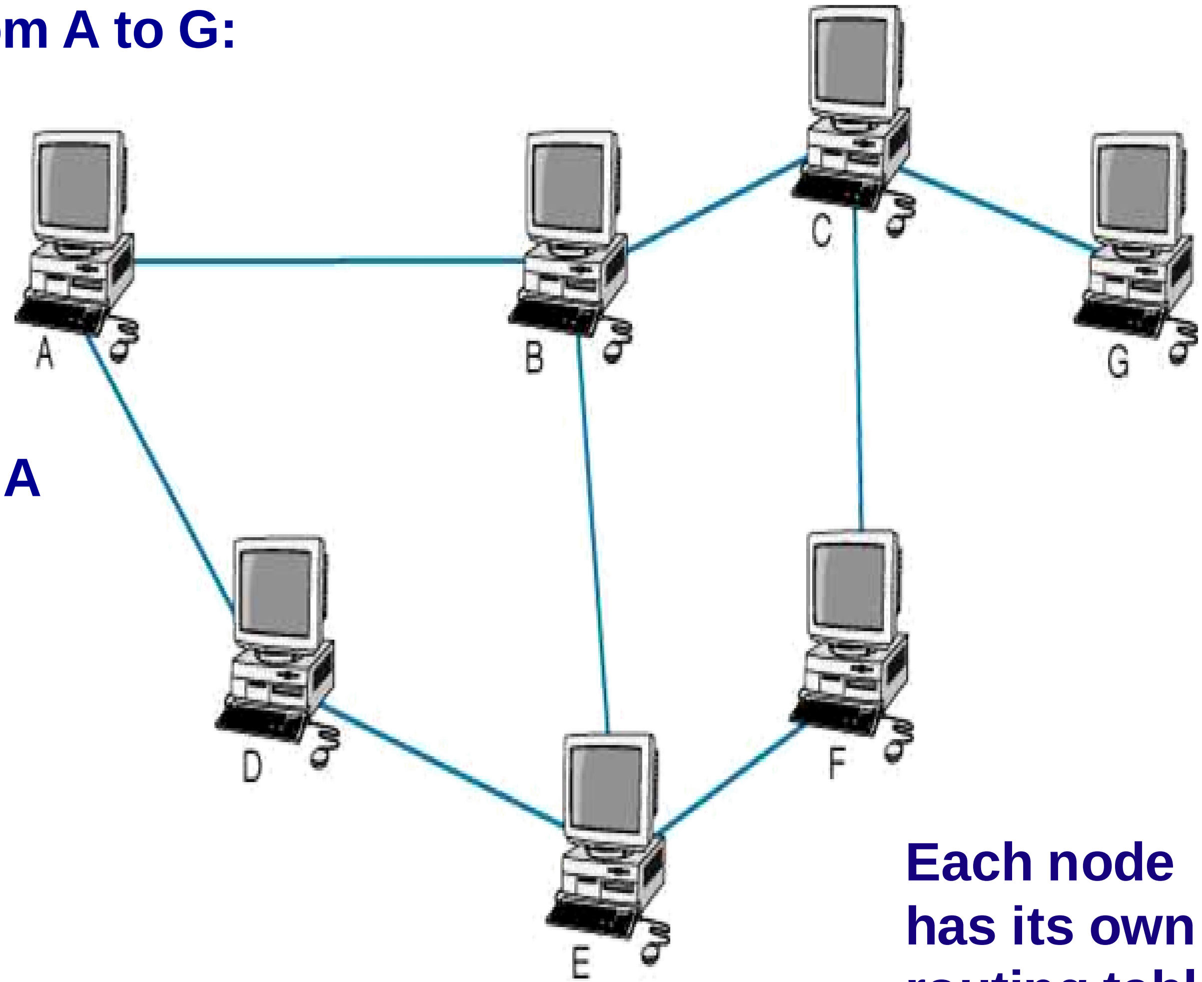
- Process of identifying what path to have a packet take through a network from sender to receiver
- Routing Tables
  - Used to make routing decisions
  - Shows which path to send packets on to reach a given destination
  - Kept by computers making routing decisions
- Routers
  - Special purpose devices used to handle routing decisions on the Internet
  - Maintain their own routing tables

Dest.	Next
B	B
C	B
D	D
E	D
F	D
G	B

Routing Example

Possible paths from A to G:

- ABCG
- ABEFCG
- ADEFCG
- ADEBCG

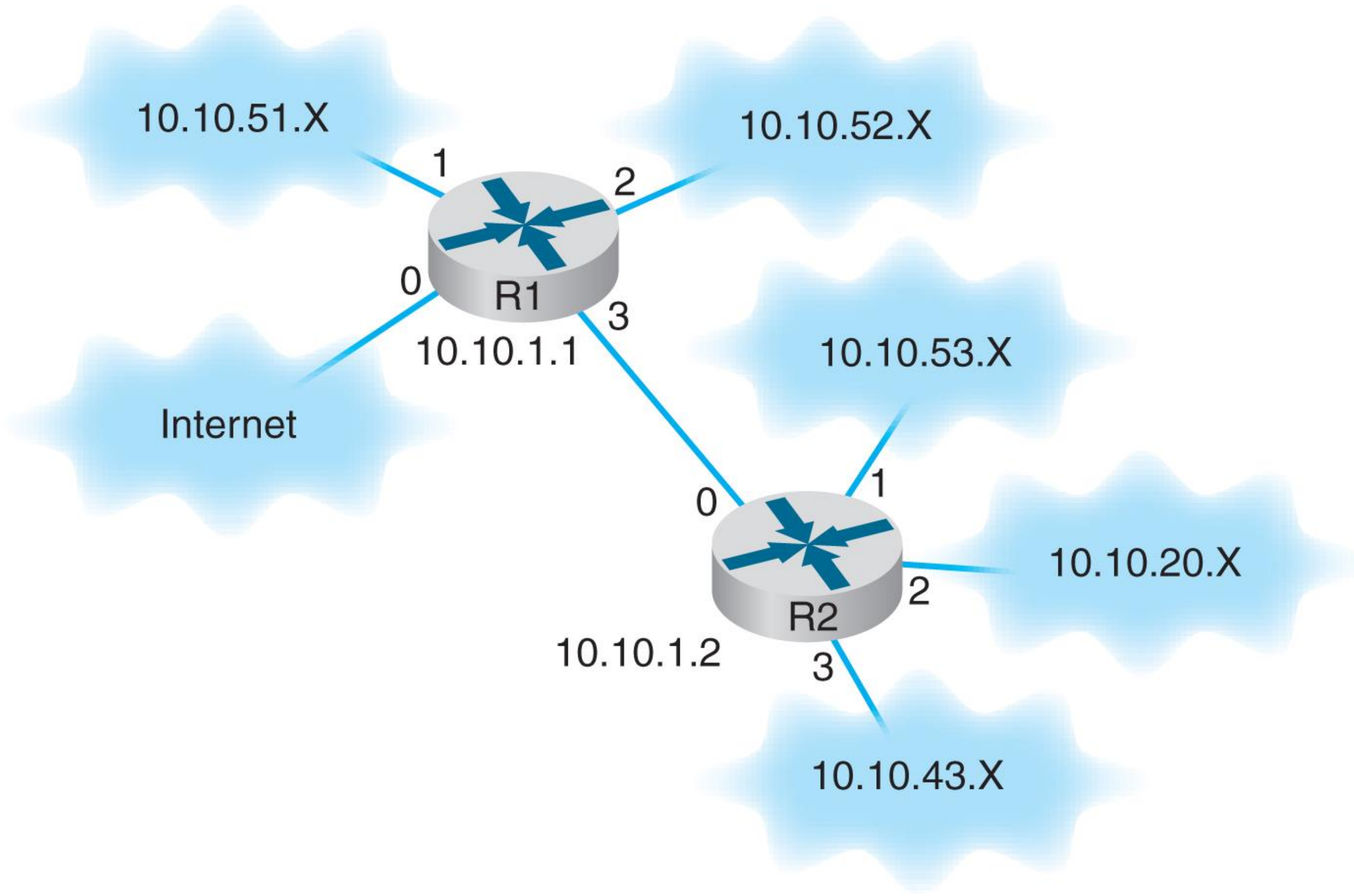


Routing Table for A

Dest.	Next
B	B
C	B
D	D
E	D
F	D
G	B

Each node has its own routing table

Routing



10.10.51.X
10.10.52.X
1
2
0
R1
3
10.10.1.1
Internet
10.10.53.X
0
1
R2
2
10.10.20.X
10.10.1.2
3
10.10.43.X

Routing

Router R1's Routing Table

Network Address	Interface
10.10.51.0 to 10.10.51.255	1
10.10.52.0 to 10.10.52.255	2
10.10.53.0 to 10.10.53.255	3
10.10.20.0 to 10.10.20.255	3
10.10.43.0 to 10.10.43.255	3
10.10.1.2	3
All other addresses	0

Router R2's Routing Table

Network Address	Interface
10.10.1.1	0
10.10.53.0 to 10.10.53.255	1
10.10.20.0 to 10.10.20.255	2
10.10.43.0 to 10.10.43.255	3
All other addresses	0

Types of Routing

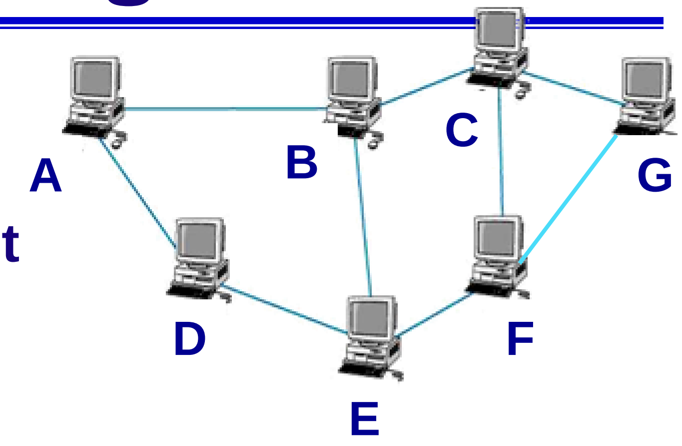
- Centralized routing
  - Decisions made by one central computer
  - Used on small, mainframe-based networks
- Decentralized routing
  - Decisions made by each node independently of one another
  - Information needs to be exchanged to prepare routing tables
  - Used by the Internet

Types of Decentralized Routing

- **Static routing:**
  - Uses fixed routing tables developed by network managers
    - Each node has its own routing table
    - Changes when computers added or removed
  - Used on relatively simple networks with few routing options that rarely change
- **Dynamic routing or Adaptive routing:**
  - Uses routing tables at each node that are updated dynamically
  - Based on routing condition information exchanged between routing devices

Dynamic Routing Algorithms

- Distance Vector
  - Uses the least number of hops to decide how to route a packet
  - Used by Routing Information Protocol (RIP)



Ex: From A to G → ABCG

- Link State
  - Uses a variety of information types to decide how to route a packet (more sophisticated)
    - e.g., number of hops, congestion, speed of circuit
  - Links state info exchanged periodically by each node to keep every node in the network up to date
  - Provides more reliable, up to date paths to destinations
  - Used by Open Shortest Path First (OSPF)

Routing Protocols

- Used to exchange info among nodes for building and maintaining routing tables
- Autonomous System (AS)
  - A network operated by an organization (e.g., Indiana U.)
  - Protocols classified based on autonomous systems
- Types of Routing Protocols
  - Interior routing protocols (RIP, OSPF, EIGRP, ICMP)
    - Operate within a network (autonomous system)
    - Provide detailed info about each node and paths
  - Exterior routing protocols (BGP)
    - Operate between networks (autonomous systems)

Routing Information Protocol (RIP)

- A dynamic distance vector interior routing protocol
- Once popular on Internet; now used on simple networks
- Operations:
  - Manager builds a routing table by suing RIP
  - Routing tables broadcast periodically (every minute or so) by all nodes
  - When a new node added, RIP counts number of hops between computers and updates routing tables

Open Shortest Path First (OSPF)

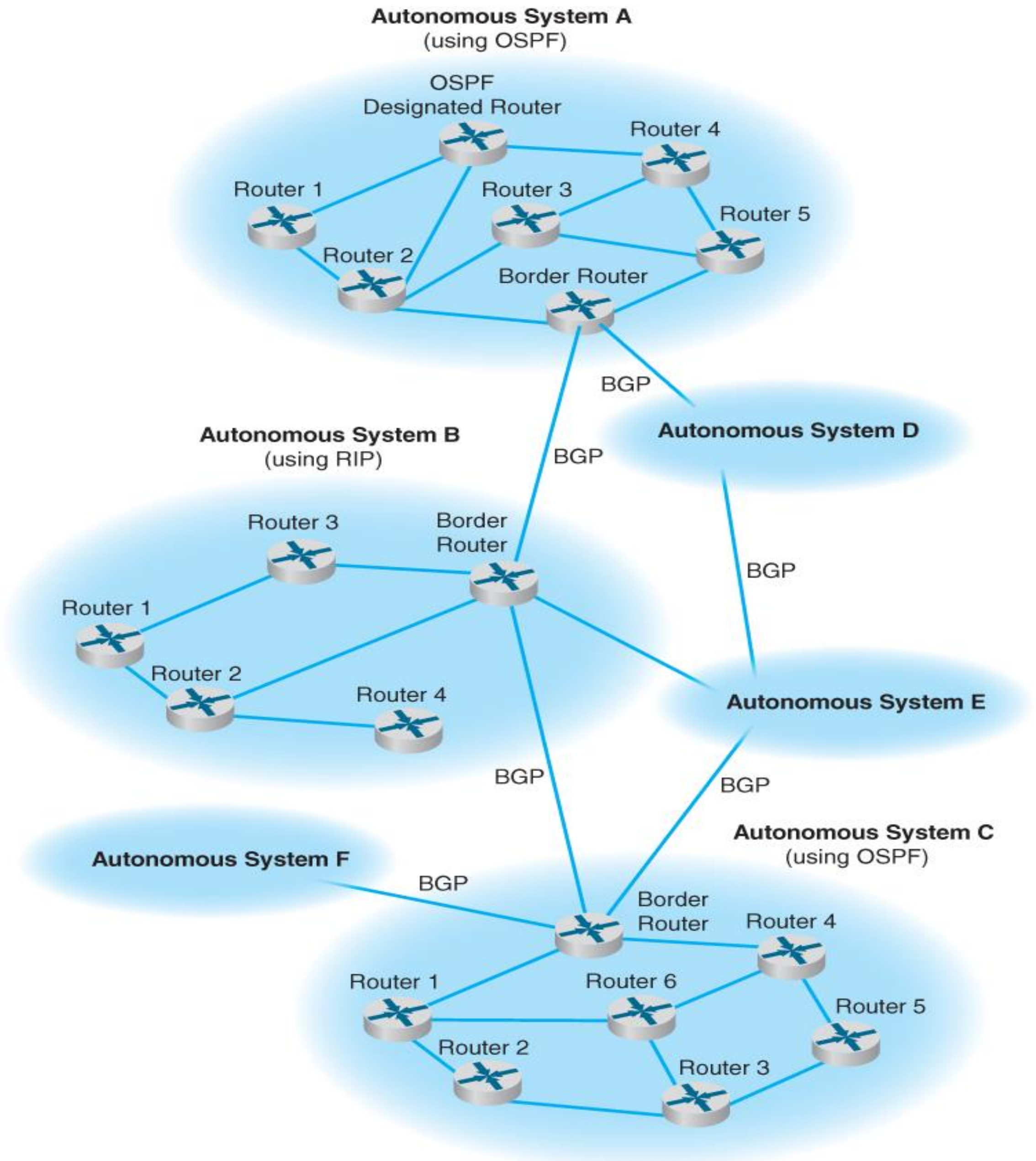
- A dynamic link state interior routing protocol
- Became more popular on Internet
  - More reliable paths
    - Incorporates traffic and error rate measures
  - Less burdensome to the network
    - Only the updates sent (not entire routing tables) and only to other routers (no broadcasting)

Other Interior Routing Protocols

- **Enhanced Interior Gateway Routing Protocol (EIGRP)**
  - A dynamic link state protocol (developed by Cisco)
  - Records transmission capacity, delay time, reliability and load for all paths
  - Keeps the routing tables for its neighbors and uses this information in its routing decisions as well
- **Internet Control Message Protocol (ICMP)**
  - Simplest and most basic
  - An error reporting protocol (report routing errors to message senders)
  - Limited ability to update routing tables

Exterior Routing Protocols

- Border Gateway Protocol (BGP)
  - Used to exchange routing info between autonomous systems
  - Based on a dynamic distance vector algorithm
  - Far more complex than interior routing protocols
  - Provide routing info only on selected routes (e.g., preferred or best route)
    - Privacy concern
    - Too many routes; can't maintain tables of every single rout



Internet Routing using BGP, OSPF and RIP

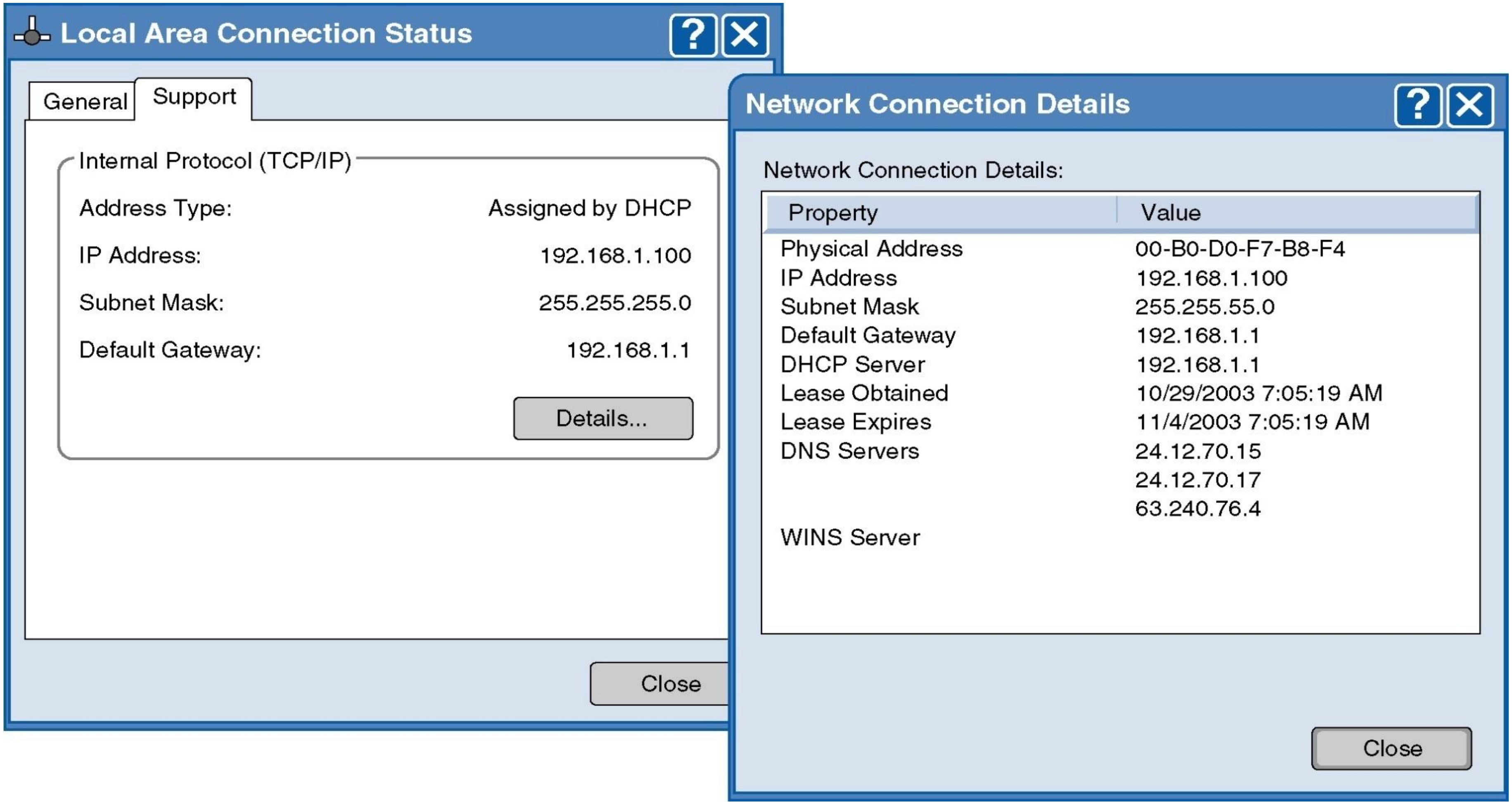
Multicasting

- **Casting**
  - ***Unicast*** **message: one computer → another computer**
  - ***Broadcast*** **message: one computer → all computers in the network**
  - ***Multicast*** **message: one computer → a group of computers (e.g., videoconference)**
- **Internet Group Management Protocol (IGMP)**
  - **Provides a way for a computer to report its multicast group membership to adjacent routers**
  - **A special IP address assigned to identify the group**
  - **Routing node sets MAC address to a matching MAC address**
  - **When multicast session ends, IGMP sends a message to the organizing computer(or router) to remove multicast group**

Sending Messages using TCP/IP

- Required Network layer addressing information
  - Computer's own IP address
  - Its subnet mask
    - To determine what addresses are part of its subnet
  - Local DNS server's IP address
    - To translate URLs into IP addresses
  - IP address of the router (gateway) on its subnet
    - To route messages going outside of its subnet
- Address information is obtained from a configuration file or provided by a DHCP server
  - Servers also need to know their own application layer addresses (domain names)

TCP/IP Configuration Information

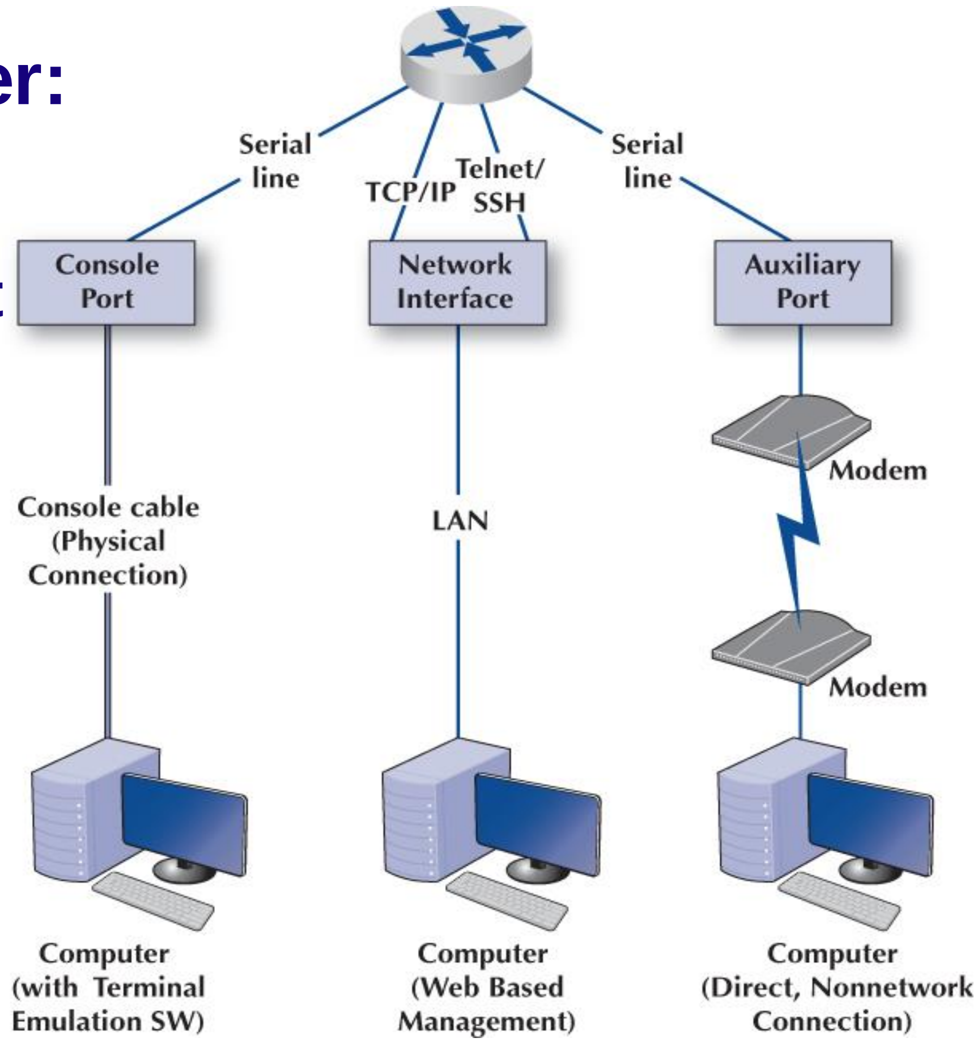


Routers

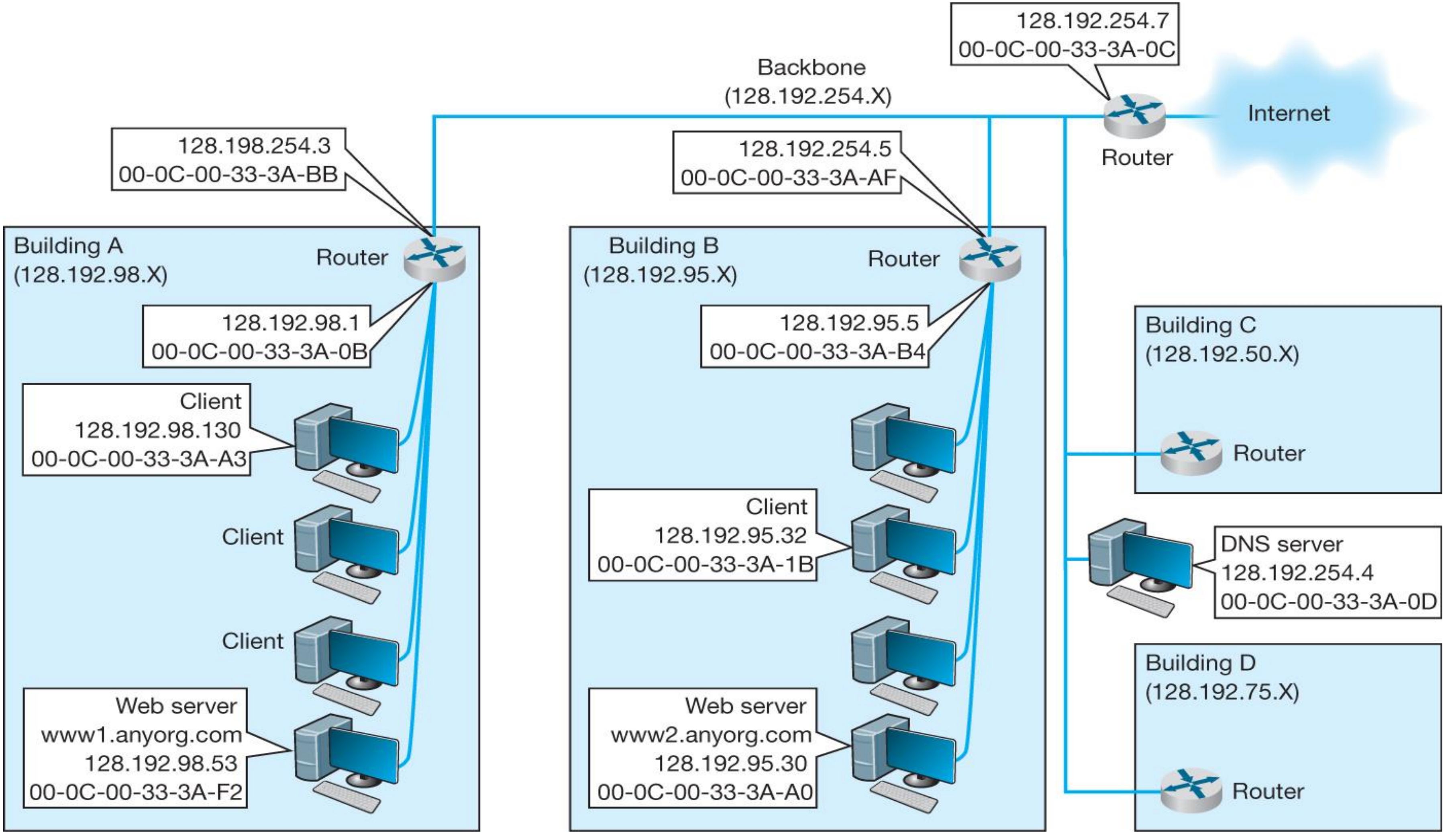
- **Functions:**
  - Determine path
  - Transmit packets
  - Support communication between variety of devices and protocols
- **Contain:**
  - CPU, memory, ports/interfaces, OS
- **Don’t contain:**
  - Disk drives, monitor, keyboard, mouse, etc

Routers

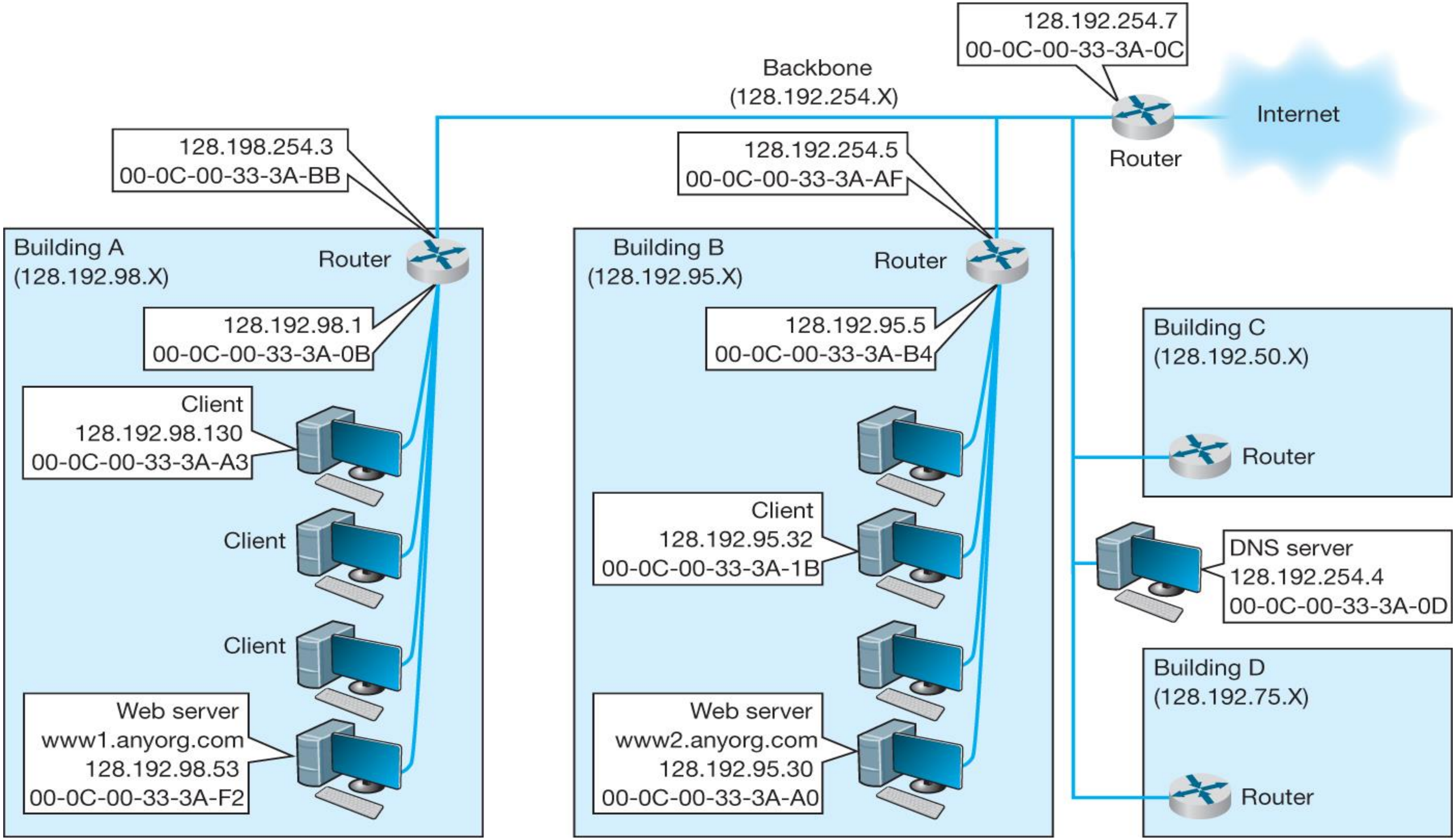
- **Connecting to a router:**
  - Console port
  - Network interface port
  - Auxiliary port
- **ACL**
  - Access Control List



5.6 TCP/IP Example



5.6 TCP/IP Example



Case 1a: Known Address, Same Subnet

- **Case:**
  - A Client (128.192.98.130) requests a Web page from a server (www1.anyorg.com)
  - Client knows the server's IP and Ethernet addresses
- **Operations (performed by the client)**
  - Prepare HTTP packet and send it to TCP
  - Place HTTP packet into a TCP packet and sent it to IP
  - Place TCP packet into an IP packet, add destination IP address, 128.192.98.53
  - Use its subnet mask to see that the destination is on the same subnet as itself
  - Add server's Ethernet address into its destination address field, and send the frame to the Web server

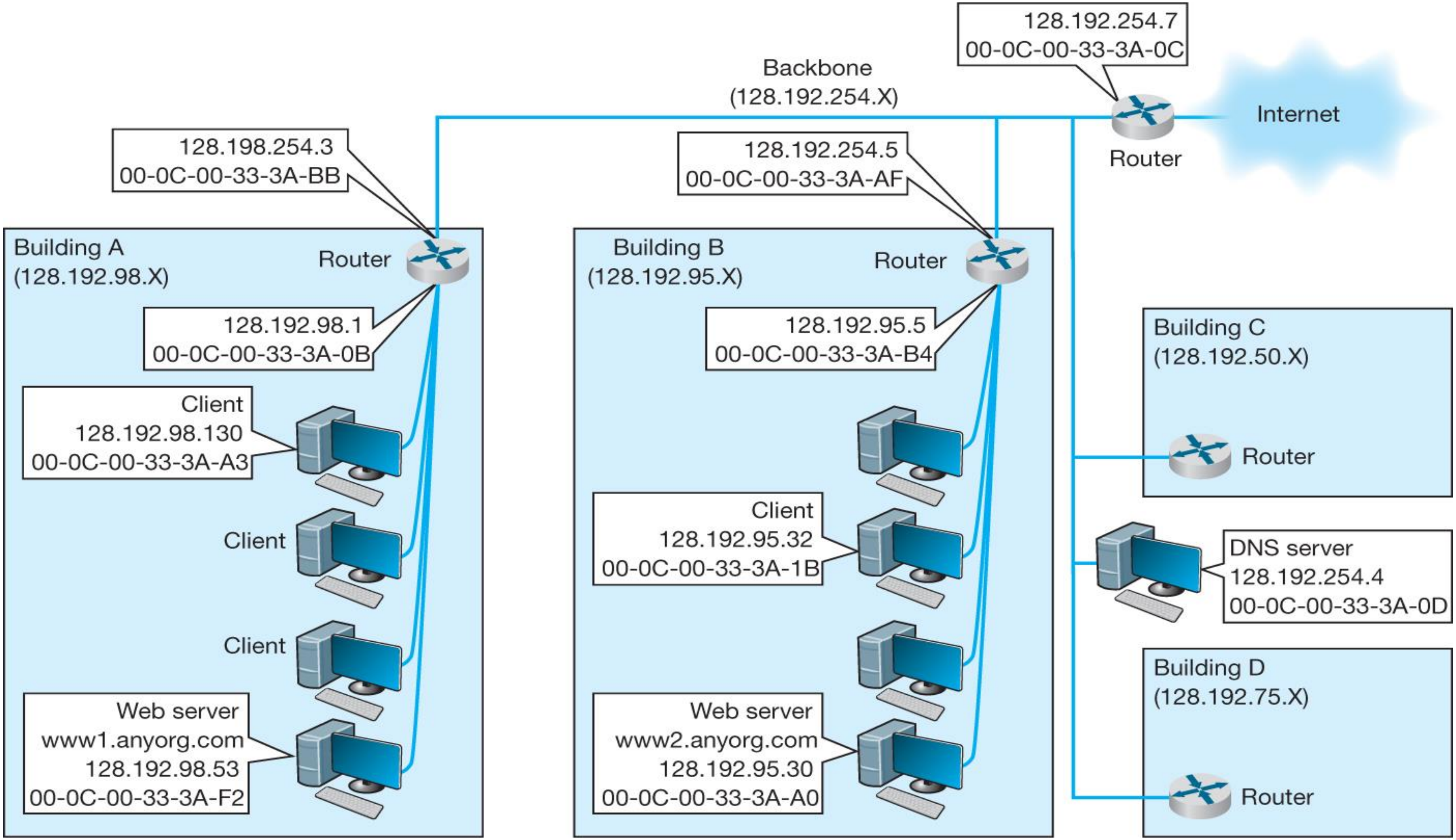
Case 1b: HTTP response to client

- Operations (performed by the server)
  - Receive Ethernet frame, perform error checking and send back an ACK
  - Process incoming frame successively up the layers (data link, network, transport and application) until the HTTP request emerges
  - Process HTTP request and sends back an HTTP response (with requested Web page)
  - Process outgoing HTTP response successively down the layers until an Ethernet frame is created
  - Send Ethernet frame to the client
- Operations (performed by the client)
  - Receive Ethernet frame and process it successively up the layers until the HTTP response emerges at browser

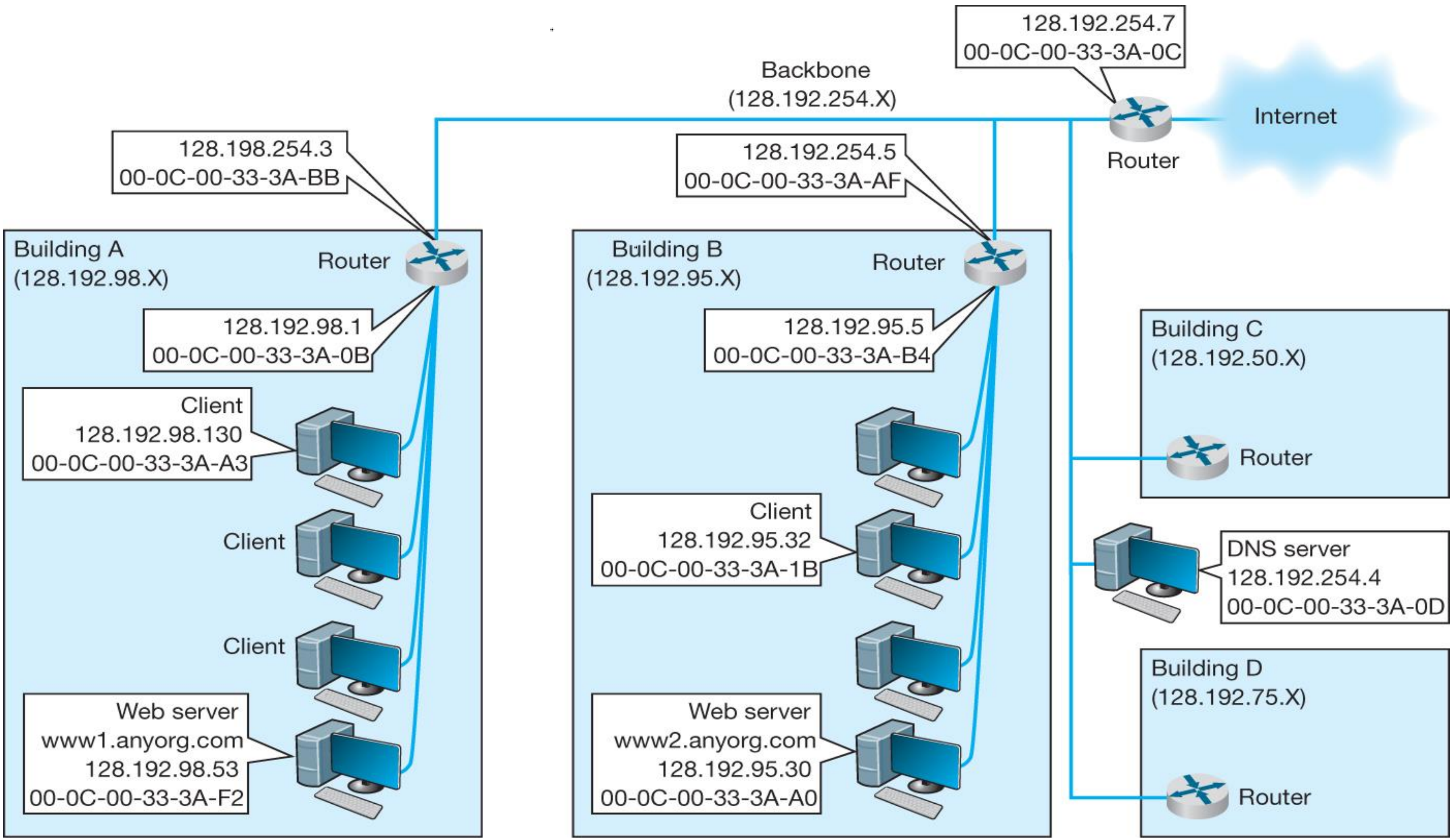
Case 2: Known Address, Different Subnet

- Similar to Case 1a
- Differences
  - Use subnet mask to determine that the destination is NOT on the same subnet
  - Send outgoing frames to the local subnet's GW
  - Local gateway operations
    - Receive the frame and remove the Ethernet header
    - Determine the next node (via Router Table)
    - Make a new frame and send it to the destination GW
  - Destination gateway operations
    - Remove the header, determine the destination (by destination IP address)
    - Place the IP packet in a new Ethernet frame and send it to its final destination.

5.6 TCP/IP Example



5.6 TCP/IP Example



Case 3: Unknown Address

- Operations (by the host)
  - Determine the destination IP address
    - Send a UDP packet to the local DNS server
    - Local DNS server knows the destination host's IP address
      - Sends a DNS response back to the sending host
    - Local DNS server does not know the destination IP address
      - Send a second UDP packet to the next highest DNS host, and so on, until the destination host's IP address is determined
  - Follow steps in Case 2

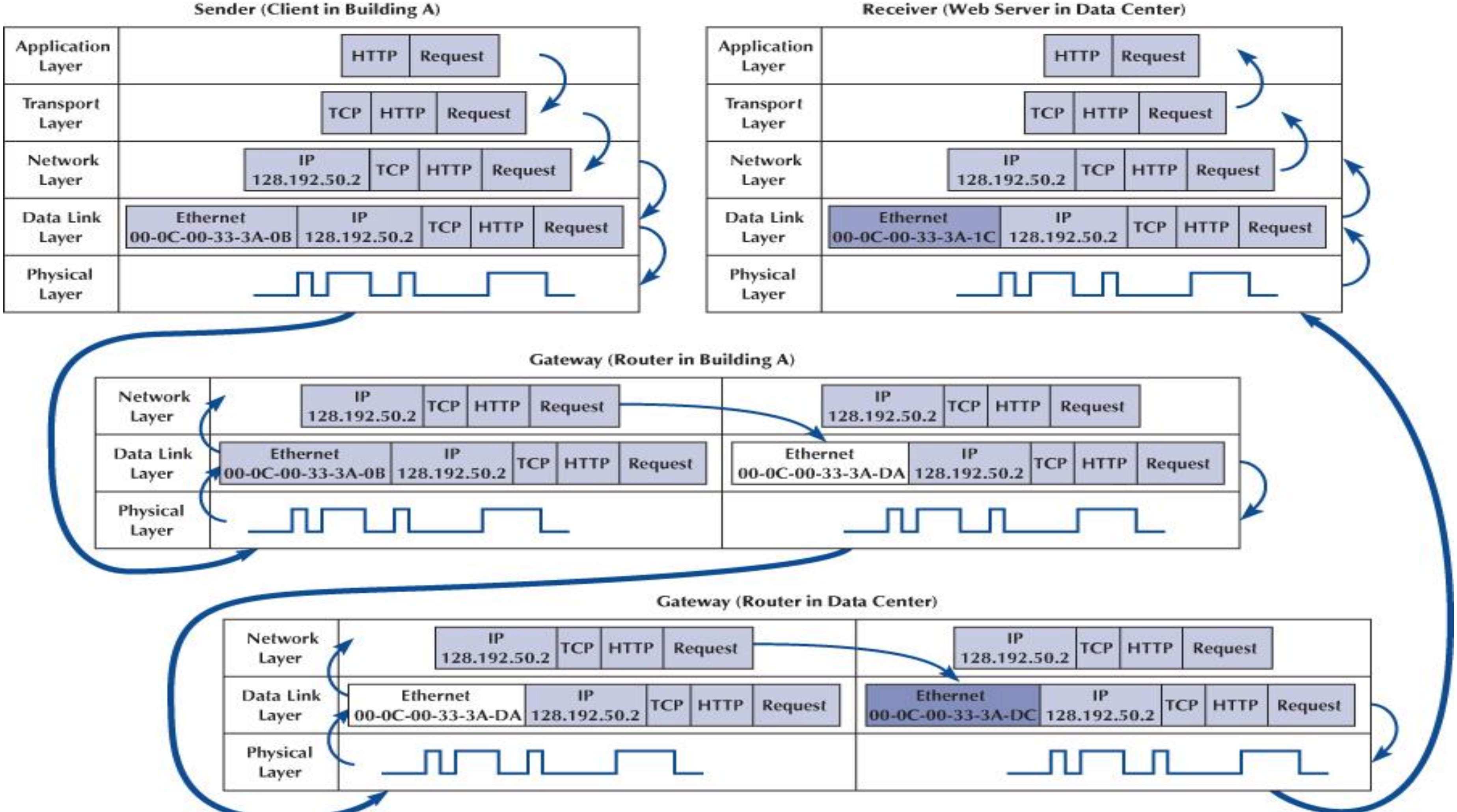
TCP Connections

- Before any data packet is sent, a connection is established
  - Use SYN packet to establish connection
  - Use FIN packet to close the connection
- Handling of HTTP packets
  - Old version:
    - a separate TCP connection for each HTTP Request
  - New version:
    - Open a connection when a request (first HTTPP Request) send to the server
    - Leave the connection open for all subsequent HTTP requests to the same server
    - Close the connection when the session ends

TCP/IP and Layers

- Host Computers
  - Packets move through all layers
- Gateways, Routers
  - Packet moves from Physical layer to Data Link Layer through the network Layer
- At each stop along the way
  - Ethernet packets is removed and a new one is created for the next node
  - IP and above packets never change in transit (created by the original sender and destroyed by the final receiver)

Message Moving Through Layers



5.7 Implications for Cyber Security

- **TCP three-way handshake can lead to DOS, Denial-Of-Service, attack**
  - **Server will keep memory reserved for false connections for awhile.**
    - **Legitimate requests won’t get response due to lack of memory**
- **IP address can expose the information of user and sell to the third entities**
  - **geographical location, city or area.**
  - **OS, browser version, time zone, etc.**

IPv6 (Internet Protocol version 6)

- The main improvement
  - IPv6 supports 128-bit versus 32-bit address in IPv4. Networked devices, such as mobile phone and mobile electronic device, have its own address.
  - With such a large address space available, IPv6 nodes can have as many universally scoped addresses as they need, and network address translation (NAT) is not required.

IPv6

IPv6 addresses

- **64-bit subnetwork prefix, and a 64-bit host.**
- **eight groups of four hexadecimal digits.**

  45a3:0db8:38e2:0000:1319:0000:0070:5700
- **0 can be omitted**

  45a3:0db8:38e2:0:1319:0:0070:5700

  45a3:0db8:38e2::1319::0070:5700

  45a3:db8:38e2::1319::70:5700

– **But, not more than one double-colon allowed**

  45a3:0db8:38e2:0000:0000:0000:0070:5700

  45a3:db8:38e2::70:5700

IPv6

Network Notation

2001:0db8:1234::/48 stands for the network with addresses 2001:0db8:1234:0000:0000:0000:0000:0000 through 2001:0db8:1234:FFFF:FFFF:FFFF:FFFF:FFFF

What is /128 ?

IPv6

IPv6 Addresses in URLs

- In a URL the IPv6-Address is enclosed in brackets.

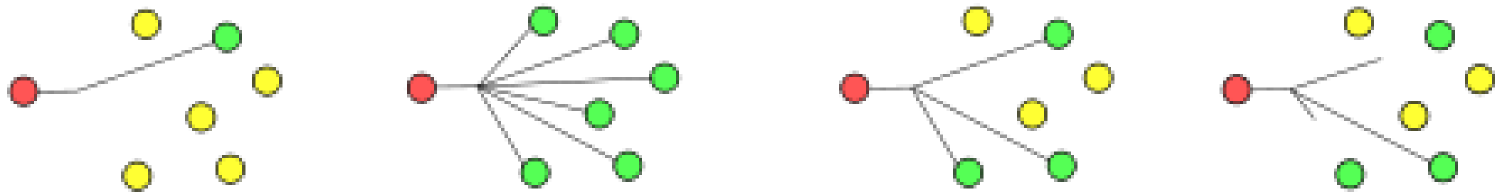
  http://[45a3:db8:38e2::1319::70:5700]/

- No confusing between IP address and port number.

  http://[45a3:db8:38e2::1319::70:5700]:8080/

IPv6

- **Multicast is part of the base protocol suite in IPv6, where IPv4 multicast is optional.**

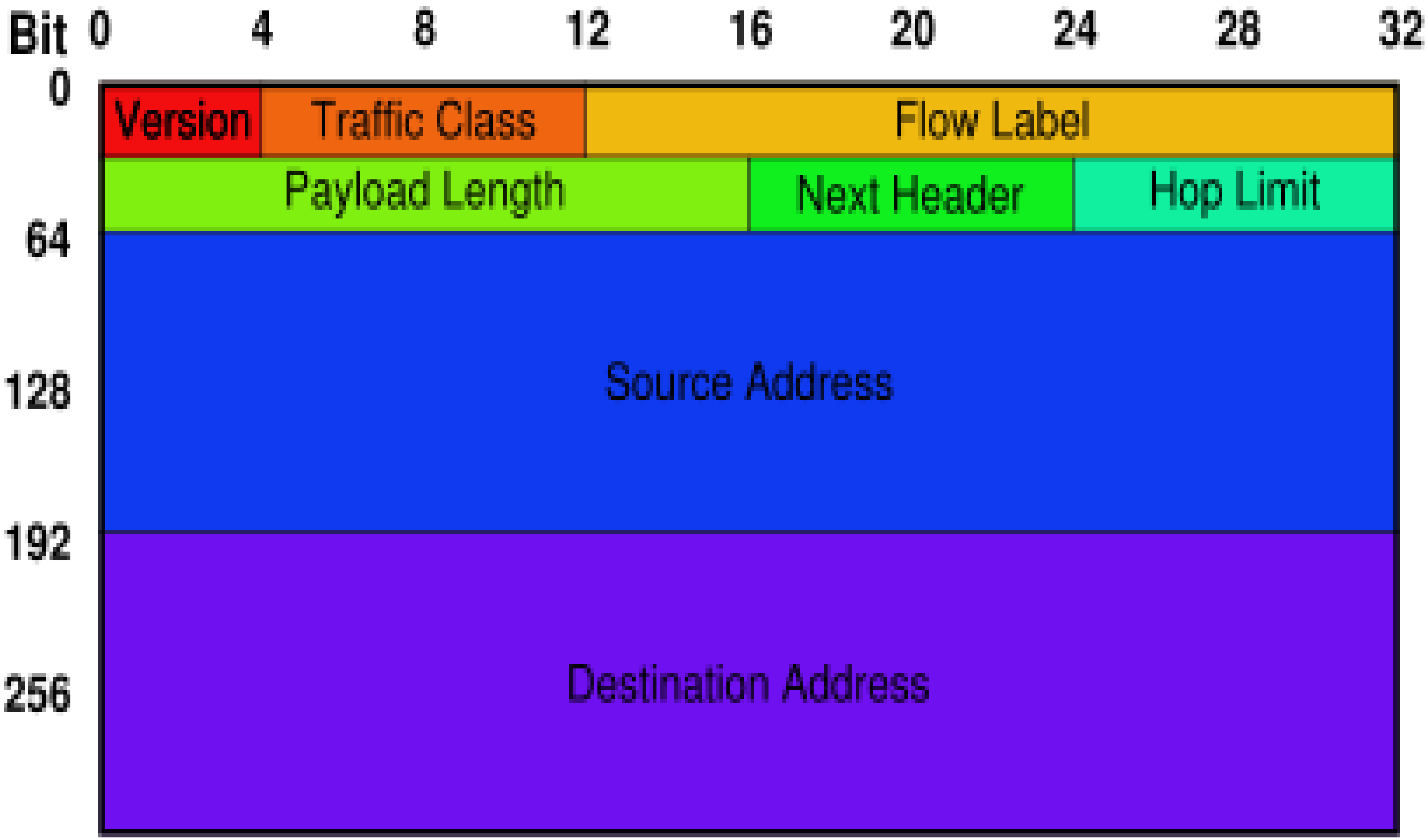


Unic ast	Broadca st (not in IPv6)	Multic ast	Anyc ast

IPv6

- **Jumbogram The packet size field of IP is 16 bits; hence, IP packets are limited to a maximum size of 64 Kbytes. An optional feature of IPv6, the *jumbo payload* option, allows the exchange of packets as large as 32 bits.**
- **Security IPsec, the protocol for IP network-layer encryption and authentication, is an integral part of the base protocol suite in IPv6; while it is optional in IPv4, but usually implemented.**

IPv6



The structure of an IPv6 packet header.

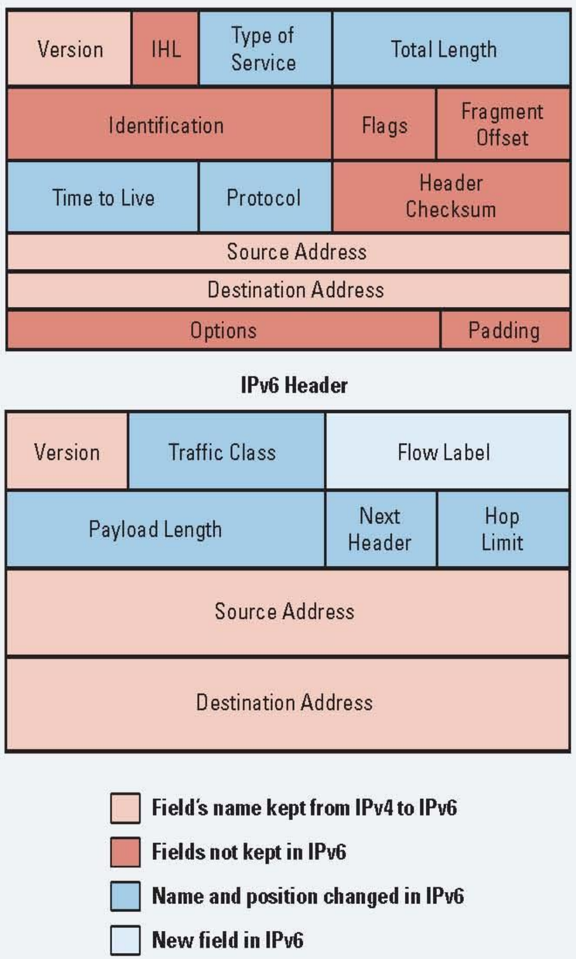
CISCO SYSTEMS

IPv6 HEADERS
AT-A-GLANCE

RFC 2460

There are several changes to the header format with IPv6. The diagrams below give a high-level view of the basic comparison between the IPv4 and IPv6 headers.

Figure 1



Streamlined

- Fragmentation fields moved out of base header
- IP options moved out of base header
- Header checksum eliminated
- Header length field eliminated
- Length field excludes IPv6 header
- Alignment changed from 32 to 64 bits

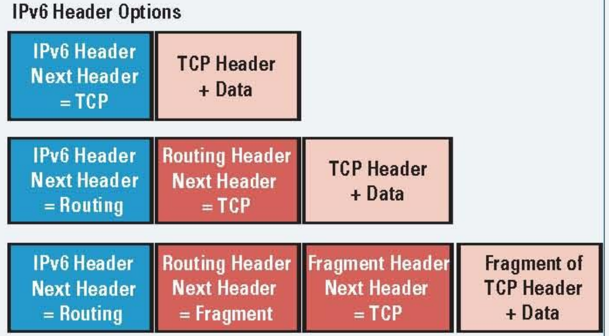
Revised

- Time to live -> hop limit
- Protocol -> next header
- Precedence and TOS -> traffic class
- Addresses increased 32 bits -> 128 bits

Extended

- Flow label field added

Figure 2

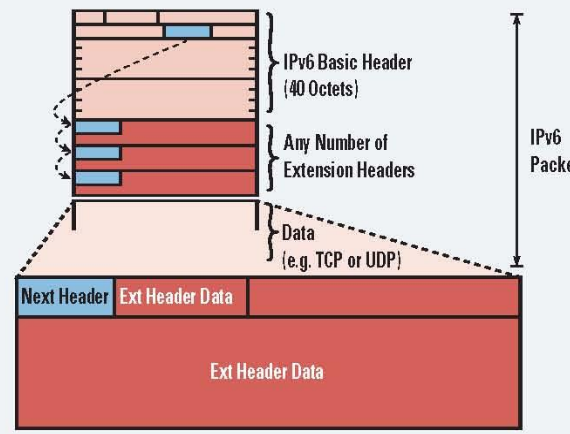


Header options processed only by node are identified in the IPv6 destination address field, except the hop-by-hop option.

Eliminates the IPv4 40-octet limit on options. In IPv6, the limit is the total packet size or max valued from the path MTU.

Headers are linked together by populating the next header (8-bit) field.

Figure 3



When more than one extension header is used in the same packet, it is recommended that those headers appear in the following order:

- IPv6 header
- Hop-by-hop options header
- Destination options header (routing header associations)
- Routing header
- Fragment header
- Authentication header
- Encapsulating security payload header
- Destination options header (options processed by final destination)
- Upper-layer header

CISCO SYSTEMS

IPv6 HEADERS AT-A-GLANCE

Table 1. Summary of Header Types and Values

Header Type	Next Header Value
Hop-by-Hop Options Header	0
Destination Option Header	60
Routing Header	43
Fragment Header	44
Authentication Header (RFC 1826) and ESP Header (RFC 1827)	51
Upper-Layer Header	6 (TCP) 17 (UDP)
Mobility Header	135

IPv6

- IPv6 is a conservative extension of IPv4. Most transport- and application-layer protocols need little or no change to work over IPv6.
- Applications, however, usually need small changes and a recompile in order to run over IPv6.

References

- Switching to VoIP, Ted Wallingford, O'Reilly, 2005
- www.ipv6forum.org
- www.wikipedia.com
- Voice Over IPv6, Daniel Minoli, Newnes (Elsevier) 2006